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Electrical cable passthrough for photovoltaic systems

Abstract

A system comprises a passthrough installed on a roof deck. The passthrough includes a base with an aperture, and a lid attached to the base. The aperture of the base is configured to align with an aperture formed within the roof deck. The system includes at least one cable and a grommet installed on the at least one cable. The lid and the base are configured to compress the grommet around the at least one cable.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a Section 111(a) application relating to and claiming the benefit of commonly-owned, U.S. Provisional Patent Application Ser. No. 63/383,844, filed Nov. 15, 2022, entitled “ELECTRICAL CABLE PASSTHROUGH FOR PHOTOVOLTAIC SYSTEMS,” the contents of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to passthrough devices and, more particularly, passthrough devices for electrical cables for photovoltaic systems.

BACKGROUND OF THE INVENTION

(2) Photovoltaic systems having solar modules are commonly installed on roofing of structures.

SUMMARY

(3) In some embodiments, a system includes a roof deck; a passthrough installed on the roof deck, wherein the passthrough includes a base having a first end and a second end opposite the first end, a first surface extending from the first end to the second end, a second surface opposite the first surface, and an aperture extending from the first surface to the second surface, a lid attached to the base, wherein the lid includes a first end substantially aligned with the first end of the base, and a sidewall between the base and the lid, wherein the first surface is juxtaposed with the roof deck, and wherein the aperture of the base of the passthrough is configured to align with an aperture formed within the roof deck; at least one cable, wherein the at least one cable includes a first end and a second end opposite the first end of the at least one cable; and a grommet installed on the at least one cable proximate to the first end of the at least one cable, wherein the lid and the base are configured to compress the grommet around the at least one cable when the lid is attached to the base, wherein the aperture of the base of the passthrough is sized and shaped to receive the at least one cable, and wherein the first end of the at least one cable extends outwardly from the first end of base and the first end of the lid of the passthrough.

(4) In some embodiments, the lid is removably attached to the base. In some embodiments, the system includes a photovoltaic module installed on the roof deck, wherein the passthrough is installed proximate to the photovoltaic module. In some embodiments, the sidewall includes at least one aperture, wherein the at least one aperture of the sidewall is sized and shaped to receive a corresponding one of the at least one cable. In some embodiments, the grommet includes at least one opening sized and shaped to receive a corresponding one of the at least one cable. In some embodiments, the sidewall includes a first end, wherein the first end of the sidewall is proximate to the first end of the base, and wherein the at least one aperture of the sidewall is located at the first end of the sidewall. In some embodiments, the base includes at least one first cavity, wherein the lid includes at least one second cavity, and wherein the at least one first cavity and the at least one second cavity are sized and shaped to form a corresponding one of the at least one aperture. In some embodiments, the grommet is located between the at least one first cavity and the at least one second cavity.

(5) In some embodiments, the grommet is configured to be compressed between the at least one first cavity and the at least one second cavity. In some embodiments, the at least one first cavity includes a plurality of first cavities, wherein the at least one second cavity includes a plurality of second cavities, and wherein the at least one aperture of the sidewall includes a plurality of apertures, wherein the at least one cable includes a plurality of cables. In some embodiments, the at least one opening of the grommet includes a plurality of openings, and wherein each of the plurality of openings is sized and shaped to receive a corresponding one of the plurality of cables. In some embodiments, the grommet includes a plurality of sections, wherein each of the plurality of openings is located in a corresponding one of the plurality of sections, and wherein each of the plurality of sections of the grommet is located between a corresponding one of the plurality of first cavities and a corresponding one of the plurality of second cavities. In some embodiments, the at least one cable includes an electrical connector, wherein the at least one aperture of the sidewall is sized and shaped to receive the electrical connector of the at least one cable. In some embodiments, the base of the passthrough includes a tubular member extending from the first surface, wherein the tubular member includes the aperture of the base, and wherein the aperture of the roof deck is sized and shaped to receive the tubular member.

(6) In some embodiments, the system further includes a receiver attached to the tubular member. In some embodiments, the system further includes an electrical box installed underneath the roof deck, and wherein the electrical box directly receives the tubular member.

(7) In some embodiments, a cable passthrough includes a base having a first end and a second end opposite the first end, a first surface extending from the first end to the second end, a second surface opposite the first surface, and an aperture extending from the first surface to the second surface; a lid attached to the base, wherein the lid includes a first end substantially aligned with the first end of the base; and a sidewall between the base and the lid, wherein the cable passthrough is configured to be installed on a roof deck such that the first surface is juxtaposed with the roof deck, and wherein the aperture of the base is configured to align with an aperture formed within the roof deck, wherein the cable passthrough is configured to receive at least one cable, wherein the aperture of the base is sized and shaped to receive the at least one cable, wherein the at least one cable is capable of extending outwardly from the first end of the base and the first end of the lid of the cable passthrough, and wherein the lid and the base are configured to compress a grommet installed around the at least one cable when the lid is attached to the base.

(8) In some embodiments, a system includes a roof deck; a passthrough installed on the roof deck, wherein the passthrough includes a base having a first end and a second end opposite the first end, a first surface extending from the first end to the second end, a second surface opposite the first surface, and an aperture extending from the first surface to the second surface, a lid attached to the base, and a sidewall between the base and the lid; a photovoltaic module installed on the roof deck, wherein the photovoltaic module includes a first side lap, wherein the first side lap includes an aperture; a roofing shingle installed on the roof deck, wherein the roofing shingle includes a first side lap, wherein the first side lap of the roofing shingle includes an aperture, wherein the first side lap of the roofing shingle overlays the first side lap of the photovoltaic module, wherein the aperture of the roofing shingle, the aperture of the photovoltaic module, and an aperture formed within the roof deck are aligned, and wherein the aperture of the base of the passthrough is aligned with the aperture of the roof deck, the aperture of the roofing shingle, and the aperture of the photovoltaic module; and at least one cable, wherein the at least one cable includes a first end and a second end opposite the first end of the at least one cable, and wherein the aperture of the base of the passthrough, the aperture of the roofing shingle, and the aperture of the photovoltaic module, are sized and shaped to receive the at least one cable, and wherein the first end of the at least one cable extends outwardly from the first end of base and the first end of the lid of the passthrough.

(9) In some embodiments, the system includes a first flashing member, wherein the first flashing member is between the passthrough and the first side lap of the roofing shingle. In some

embodiments, the system includes a second flashing member, wherein the second flashing member is between the first side lap of the roofing shingle and the first side lap of the photovoltaic module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1 and 2 illustrate some embodiments of an electrical cable passthrough;
- (2) FIGS. 3A and 3B illustrate some embodiments of an electrical cable passthrough, a photovoltaic module, and a roofing shingle installed on a roof deck;
- (3) FIGS. 3C and 3D illustrate some embodiments of a roofing shingle and a photovoltaic module shown in FIG. 3A and FIG. 4;
- (4) FIGS. 4 through 4D illustrate some embodiments of an electrical cable passthrough and components thereof, a photovoltaic module, and a roofing shingle installed on a roof deck; and
- (5) FIGS. 5 and 6 illustrate some embodiments of an electrical cable passthrough installed on a roof deck.

DETAILED DESCRIPTION

- (6) Referring to FIGS. 1 and 2, in some embodiments, an electrical cable passthrough **10** (hereinafter “passthrough **10**”) is configured to be installed on a roof deck **100** of a structure. In some embodiments, the passthrough **10** is a component of a photovoltaic system. In some embodiments, the passthrough **10** includes a base **12** and a lid **14**. In some embodiments, the lid **14** is configured to be attached to the base **12**. In some embodiments, the lid **14** is configured to be removably attached to the base **12**.
- (7) In some embodiments, the base **12** includes a first end **11** and a second end **13** opposite the first end **11**. In some embodiments, the base **12** includes a base plate **16**. In some embodiments, the base plate **16** includes a first surface **18** and a second surface **20** opposite the first surface **18**. In some embodiments, the first surface **18** extends from the first end **11** to the second end **13**. In some embodiments, the second surface **20** extends from the first end **11** to the second end **13**. In some embodiments, the base plate **16** has a circular shape. In some embodiments, the base plate **16** is substantially circular in shape. In some embodiments, the base plate **16** has a square shape. In some embodiments, the base plate **16** has a rectangular shape. In some embodiments, the base plate **16** has a triangular shape. In some embodiments, the base plate **16** has an elliptical shape. In some embodiments, the base plate **16** has a polygonal shape. In some embodiments, the base plate **16** has at least one aperture **22** extending from the first surface **18** to the second surface **20**. In some embodiments, the at least one aperture **22** includes a plurality of apertures **22**. In some embodiments, each of the apertures **22** is sized and shaped to receive a fastener **24**. In some embodiments, the fastener **24** is a screw. In some embodiments, the fastener **24** is a bolt. In some embodiments, the fastener **24** is a nail. In some embodiments, the fasteners **24** are configured to mount and secure the passthrough **10** on the roof deck **100**. In some embodiments, the base plate **16** is configured to be positioned substantially flush with the roof deck **100**. In some embodiments, the base plate **16** is configured to be positioned flush with the roof deck **100**.
- (8) In some embodiments, an aperture **26** extends from the first surface **18** to the second surface **20**. In some embodiments, the aperture **26** is circular in shape. In some embodiments, the aperture **26** is square in shape. In some embodiments, the aperture **26** is rectangular in shape. In some embodiments, the aperture **26** is polygonal in shape. In some embodiments, the base **12** includes a tubular portion **28** extending from the second surface **20**. In some embodiments, the aperture **26** extends from the first surface **18** to the tubular portion **28**. In some embodiments, the tubular portion **28** includes internal threads **30**. In some embodiments, the tubular portion **28** is sized and shaped to receive a conduit **110**. In some embodiments, external threads **112** of the conduit **110** threadedly engage the internal threads **30** of the tubular portion **28**. In some embodiments, the

tubular portion **28** includes external threads and the conduit **110** includes internal threads that threadedly engage with the external threads of the tubular portion. In some embodiments, the conduit **110** is attached to the tubular portion **28** by a fastener. In some embodiments, the conduit **110** is attached to the tubular portion **28** by an adhesive. In some embodiments, the conduit **110** is attached to the tubular portion **28** by welding. In some embodiments, a seal **111** is located below the base **12**. In some embodiments, the seal **111** includes a first portion **113**. In some embodiments, the first portion **113** is located below the base **12**. In some embodiments, the first portion **113** is sized and shaped to match or substantially match the size and shape of the lower surface **15** of the base **12**. In some embodiments, the seal **111** includes a second portion **115**. In some embodiments, the second portion **115** extends from the first portion **113**. In some embodiments, the second portion **115** is located below the sidewall **32**. In some embodiments, the second portion **115** is sized and shaped to match or substantially match the size and shape of the lower surface of the sidewall **32**. In some embodiments, the seal **111** is composed of butyl, silicone, rubber, epoxy, latex, neoprene, or polyurethane foam.

(9) In some embodiments, the base **12** includes a sidewall **32**. In some embodiments, the sidewall **32** extends outwardly from the base plate **16**. In some embodiments, the sidewall **32** extends outwardly from the first surface **18** of the base plate **16**. In some embodiments, the sidewall **32** includes a first portion **34**. In some embodiments, the first portion **34** circumferentially surrounds the aperture **26** at the first surface **18**. In some embodiments, the first portion **34** is substantially circular in shape. In some embodiments, the sidewall **32** includes a second portion **36**. In some embodiments, the second portion **36** is adjacent the first portion **34**. In some embodiments, the second portion **36** extends from the first portion **34** to the first end **11**. In some embodiments, the second portion **36** is oblong in shape. In some embodiments, the second portion **36** is substantially rectangular in shape. In some embodiments, the first portion **34** and the second portion **36** form a keyhole shape.

(10) In some embodiments, the second portion **36** includes a bore **35**. In some embodiments, the bore **35** includes an internal thread. In some embodiments, the bore **35** is configured to receive a fastener **37**. As discussed below, in some embodiments, the fastener **37** is configured to fasten the lid **14** to the base **12**. In some embodiment, the second portion **36** includes at least one tab **39** extending outwardly proximate the second end **13**. In some embodiments, the at least one tab **39** includes a plurality of tabs **39**. As discussed below, in some embodiments, the at least one tab **39** is sized and shaped to engage a slot **58** of the lid **14**.

(11) In some embodiments, the sidewall **32** includes a first end **38**. In some embodiments, the first end **38** of the sidewall **32** is proximate to the first end **11** of the base **12**. In some embodiments, at least one first cavity **40** is located at the first end **38** of the sidewall **32**. In some embodiments, the at least one first cavity **40** includes one first cavity **40**. In some embodiments, the at least one first cavity **40** includes a plurality of first cavities **40**. In some embodiments, the plurality of first cavities **40** includes two of the first cavities **40**. In some embodiments, the plurality of first cavities **40** includes three of the first cavities **40**. In some embodiments, the plurality of first cavities **40** includes four of the first cavities **40**. In some embodiments, the plurality of first cavities **40** includes more than four of the cavities **40**. In some embodiments, each of the first cavities **40** is semi-tubular or semi-cylindrical in shape.

(12) In some embodiments, the lid **14** includes a first end **42** and a second end **44** opposite the first end **42**. In some embodiments, the lid **14** includes an outer wall **46**. In some embodiments, the outer wall **46** includes a first portion **48**. In some embodiments, the first portion **48** is substantially circular in shape. In some embodiments, the outer wall **46** includes a second portion **50**. In some embodiments, the second portion **50** is adjacent the first portion **48**. In some embodiments, the second portion **50** extends from the first portion **48** to the first end **42**. In some embodiments, the second portion **50** is oblong in shape. In some embodiments, the second portion **50** is substantially rectangular in shape. In some embodiments, the first portion **48** and the second portion **50** form a

keyhole shape. In some embodiments, the lid **14** is attached to the sidewall **32** of the base **12**. In some embodiments, the lid **14** has a size and shape that is substantially similar to the size and shape of the sidewall **32** of the base **12**. In some embodiments, the first portion **48** of the lid **14** covers the first portion **34** of the sidewall **32** of the base **12**. In some embodiments, the second portion **50** of the lid **14** covers the second portion **36** of the sidewall **32** of the base **12**. In some embodiments, the first end **42** of the lid **14** is substantially aligned with the first end **11** of the base **12**. In some embodiments, the first end **42** of the lid **14** is aligned with the first end **11** of the base **12**. In some embodiments, the lid **14** is hollow and with the base **12** forms an interior portion **51**. In some embodiments, the interior portion **51** is sized and shaped to receive at least one electrical cable **120**. In some embodiments, the passthrough includes a O-ring **57**. In some embodiments, the O-ring **57** is between the base **12** and the lid **14**. In some embodiments, the O-ring **57** is between the sidewall **32** of the base **12** and the outer wall **46** of the lid. In some embodiments, the O-ring **57** forms a watertight seal between the base **12** and the lid **14**. In some embodiments, the O-ring **57** is made from rubber. In some embodiments, the O-ring **57** is made from styrene-butadiene or styrene-butadiene rubber (SBR). In some embodiments, the O-ring **57** is made from nitrile butadiene rubber (NBR). In some embodiments, the O-ring **57** is made from EPDM rubber (ethylene propylene diene monomer rubber). In some embodiments, the O-ring **57** is made from plastic. In some embodiments, the O-ring **57** is made from a polymer. In some embodiments, the O-ring **57** is made from silicone. In some embodiments, the O-ring **57** is made from polytetrafluoroethylene (PTFE). In some embodiments, the O-ring **57** is made from fluorosilicone (FVMQ). In some embodiments, the O-ring **57** is made from polyurethane.

(13) In some embodiments, the lid **14** includes an inner wall **53** and a receptacle **55** extending from the inner wall **53**. In some embodiments, the receptacle **55** includes an internal thread. In some embodiments, the bore **35** and the receptacle **55** are configured to receive the fastener **37** to fasten the lid **14** to the base **12**. In some embodiments, the fastener **37** is a screw. In some embodiments, the fastener **37** is a compression screw. In some embodiments, the fastener **37** is bolt. In some embodiments, the fastener **37** is a threaded rod.

(14) In some embodiments, at least one second cavity **52** is located at the first end **42** of the lid **14**. In some embodiments, the at least one second cavity **52** includes a single cavity **52**. In some embodiments, the at least one second cavity **52** includes a plurality of second cavities **52**. In some embodiments, the plurality of second cavities **52** includes two of the second cavities **52**. In some embodiments, the plurality of second cavities **52** includes three of the second cavities **52**. In some embodiments, the plurality of second cavities **52** includes four of the second cavities **52**. In some embodiments, the plurality of second cavities **52** includes more than four of the second cavities **52**. In some embodiments, each of the second cavities **52** is semi-tubular or semi-cylindrical in shape.

(15) In some embodiments, at least one aperture **54** is located at the first end **38** of the sidewall **32**. In some embodiments, the at least one aperture **54** of the sidewall **32** is sized and shaped to receive a corresponding one of the at least one electrical cable **120**. In some embodiments, the at least one first cavity **40** of the base **12** and the at least one second cavity **52** of the lid **14** are sized and shaped to form a corresponding one of the at least one aperture **54**. In some embodiments, the at least one aperture **54** includes a plurality of apertures **54**. In some embodiments, the plurality of first cavities **40** of the base **12** and the plurality of second cavities **52** of the lid **14** are sized and shaped to form a corresponding one of the a plurality of the apertures **54**. In some embodiments, each of the plurality of the apertures **54** is sized and shaped to receive a corresponding one of the a plurality of the electrical cables **120**. In some embodiments, the plurality of the apertures **54** includes two of the apertures **54**. In some embodiments, the plurality of the electrical cables **120** includes two of the electrical cables **120**. In some embodiments, the plurality of the apertures **54** includes three of the apertures **54**. In some embodiments, the plurality of the electrical cables **120** includes three of the electrical cables **120**. In some embodiments, the plurality of the apertures **54** includes four of the apertures **54**. In some embodiments, the plurality of the electrical cables **120** includes four of the

electrical cables **120**. In some embodiments, the plurality of the apertures **54** includes more than four of the apertures **54**. In some embodiments, the plurality of the electrical cables **120** includes more than four of the electrical cables **120**. In some embodiments, the at least one aperture **54** is circular in shape. In some embodiments, the at least one aperture **54** is square in shape.

(16) In some embodiments, the lid **14** includes at least one slot **58** at the second end **44** thereof that is sized and shaped to receive a corresponding one of the at least one tab **39** of the base **12**. In some embodiments, the at least one slot **58** includes a plurality of slots **58**. In some embodiments, the at least one tab **39** includes a plurality of tabs **39**. In some embodiments, each of the tabs **39** is sized and shaped to engage a corresponding one of the slots **58** to removably secure the lid **14** to the base **12**.

(17) In some embodiments, one or both of the base **12** and the lid **14** are made from plastic. In some embodiments, the base **12** and the lid **14** are made from polypropylene. In some embodiments, the base **12** and the lid **14** are made from a polymer blend with polypropylene as a base resin. In another embodiment, the base **12** and the lid **14** are made from thermoplastic polyolefin (TPO). In another embodiment, the TPO is a modified TPO including fiberglass and/or other filler material. In another embodiment, the base **12** and the lid **14** are made from metal. In some embodiments, the base **12** and the lid **14** are made of aluminum. In another embodiment, the base **12** and the lid **14** are made of stainless steel.

(18) In some embodiments, the at least one electrical cable **120** includes the at least one electrical connector **122**. In some embodiments, the at least one electrical connector **122** is a bulkhead (i.e., panel mount) connector. In some embodiments, the at least one electrical connector **122** is electrically connected to jumper modules, jumper cables, junction boxes, or module leads.

(19) In some embodiments, a grommet **60** is installed on the at least one electrical cable **120**. In some embodiments, a grommet **60** is installed proximate to a first end **121** of the at least one electrical cable **120**. In some embodiments, the grommet **60** includes at least one opening **62**. In some embodiments, the at least one opening **62** is sized and shaped to receive a corresponding one of the at least one electrical cable **120**. In some embodiments, the grommet **60** is installed and located between the at least one first cavity **40** of the base **12** and the at least one second cavity **52** of the lid **14**. In some embodiments, the grommet **60** is configured to be compressed between the at least one first cavity **40** of the base **12** and the at least one second cavity **52** of the lid **14** when the lid **14** is attached to the base **12**. In some embodiments, when the fastener **37** is tightened to attached the lid **14** to the base **12**, the grommet **60** is compressed. In some embodiments, a user may selectively tighten or loosen the fastener **37** to control the amount of force on and compression of the grommet **60**. In some embodiments, the compression of the grommet **60** forms a watertight seal with the electrical cables **120** to inhibit the entry of water or moisture in the interior portion **51** of the passthrough **10**. In some embodiments, the grommet **60** is elastically deformed. In some embodiments, the grommet **60** is made from rubber. In some embodiments, the grommet **60** is made from styrene-butadiene or styrene-butadiene rubber (SBR). In some embodiments, the grommet **60** is made from nitrile butadiene rubber (NBR). In some embodiments, the grommet **60** is made from EPDM rubber (ethylene propylene diene monomer rubber). In some embodiments, the grommet **60** is made from plastic. In some embodiments, the grommet **60** is made from a polymer. In some embodiments, the grommet **60** is made from silicone. In some embodiments, the grommet **60** is made from polytetrafluoroethylene (PTFE). In some embodiments, the grommet **60** is made from fluorosilicone (FVMQ). In some embodiments, the grommet **60** is made from polyurethane.

(20) In some embodiments, the at least one opening **62** of the grommet **60** includes a plurality of openings **62**. In each of the plurality of openings **62** is sized and shaped to receive a corresponding one of the plurality of the electrical cables **120**. In some embodiments, the grommet **60** includes a plurality of sections **64**. In some embodiments, each of the plurality of sections **64** has a tubular shape. In some embodiments, the plurality of sections **64** are integral. In some embodiments, the

grommet **60** has a figure eight shape. In some embodiments, each of the plurality of openings **62** is located in a corresponding one of the plurality of sections **64**. In some embodiments, each of the plurality of sections **64** of the grommet **60** is located between a corresponding one of the plurality of first cavities **40** of the base **12** and a corresponding one of the plurality of second cavities **52** of the lid **14**. In some embodiments, the plurality of openings **62** is arranged in a linear arrangement. In some embodiments, the grommet **60** has a figure eight shape.

(21) In some embodiments, the passthrough **10** has a thickness of 1 inch to 10 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 9 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 8 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 7 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 6 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 5 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 4 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 3 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 2 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 6 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 5 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 4 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 3 inches.

(22) In some embodiments, the passthrough **10** has a thickness of 3 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 6 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 5 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 4 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 6 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 5 inches.

(23) In some embodiments, the passthrough **10** has a thickness of 5 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches to 6 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 7 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 7 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 7 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 8 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 8 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 9 inches to 10 inches.

(24) In some embodiments, the passthrough **10** has a thickness of 1 inch. In some embodiments, the passthrough **10** has a thickness of 2 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches. In some embodiments, the

passthrough **10** has a thickness of 6 inches. In some embodiments, the passthrough **10** has a thickness of 7 inches. In some embodiments, the passthrough **10** has a thickness of 8 inches. In some embodiments, the passthrough **10** has a thickness of 9 inches. In some embodiments, the passthrough **10** has a thickness of 10 inches.

(25) Referring to FIGS. 3A and 3B, in some embodiments, the passthrough **10** is installed on the roof deck **100**. In some embodiments, the roof deck **100** includes an aperture **101**. Referring to FIGS. 3A through 3D, in some embodiments, at least one roofing shingle **200a** (also referred to as a roofing module) is installed on the roof deck **100**. In some embodiments, the roofing shingle **200a** does not include a solar cell, a photovoltaic cell, or any electrical device that converts the energy of light into electricity. In some embodiments, the roofing shingle **200a** does not include any electrically active components. In some embodiments, the roofing shingle **200a** is a cuttable roofing shingle. In some embodiments, the roofing shingle **200a** is a nailable roofing shingle. In some embodiments, the roofing shingle **200a** is attached to the roof deck **100** by at least one fastener or a plurality of fasteners. In some embodiments, the fasteners include nails, rivets, screws or staples, or a combination thereof. In some embodiments, the roofing shingle **200a** is textured. In some embodiments, the roofing shingle **200a** includes a printed pattern **203**. In some embodiments, the printed pattern **203** includes a depiction of solar cells. In some embodiments, the at least one roofing shingle **200a** includes a plurality of roofing shingles **200a**. In some embodiments, each of the roofing shingles **200a** includes a first end **202a** and a second end **204a** opposite the first end **202a**. In some embodiments, each of the roofing shingles **200a** includes a first side lap **206a** at the first end **202a**. In some embodiments, each of the roofing shingles **200a** includes a second side lap **208a** at the second end **204a**. In some embodiments, the first side lap **206a** of the roofing shingle **200a** includes an aperture **210a** extending from an upper surface to a lower surface thereof. In some embodiments, the roofing shingle **200a** includes a head lap portion **215a**. In some embodiments, the head lap portion **215a** is configured to receive one or more of the fasteners for installing the roofing shingle **200a** to the roof deck. In some embodiments, at least a portion of a first roofing shingle **200a** is configured to overlay the head lap portion **215a** of a second roofing shingle **200a**.

(26) In some embodiments, the roofing shingle **200a** includes a structure, composition and/or function of similar to those of more or one of the embodiments of the roofing modules disclosed in in U.S. application Ser. No. 17/831,307, filed Jun. 2, 2022, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2022-0393637 on Dec. 8, 2022, and owned by GAF Energy LLC; and/or U.S. application Ser. No. 18/169,718, filed Feb. 15, 2023, titled “Roofing Module System,” and published under U.S. Patent Application Publication No. 2023-0203815 on Jun. 29, 2023, and owned by GAF Energy LLC, the disclosure and contents of each of which are incorporated by reference herein in their entirety.

(27) In some embodiments, at least one photovoltaic module **200b** is installed on the roof deck **100**. In some embodiments, the photovoltaic module **200b** includes a plurality of solar cells **201**. In some embodiments, the at least one photovoltaic module **200b** includes a plurality of photovoltaic modules **200b**. In some embodiments, each of the photovoltaic modules **200b** includes a first end **202b** and a second end **204b** opposite the first end **202b**. In some embodiments, each of the photovoltaic modules **200b** includes a first side lap **206b** at the first end **202b**. In some embodiments, each of the photovoltaic modules **200b** includes a second side lap **208b** at the second end **204b**. In some embodiments, the second side lap **208b** of the photovoltaic module **200b** includes an aperture **212b** extending from an upper surface to a lower surface thereof.

(28) In some embodiments, the photovoltaic module **200b** includes a head lap portion **215b**. In some embodiments, the head lap portion **215b** is configured to receive one or more fasteners. In some embodiments, at least a portion of a first photovoltaic module **200b** is configured to overlay the head lap portion **215b** of a second photovoltaic module **200b**.

(29) In some embodiments, at least a portion of a first photovoltaic module **200b** is configured to

overlay the head lap portion **215a** of a first roofing shingle **200a**. In some embodiments, at least a portion of a first roofing shingle **200a** is configured to overlay the head lap portion **215b** of a first photovoltaic module **200b**.

(30) In some embodiments, each of the photovoltaic modules **200b** includes a structure, composition, features, components, and/or function similar to those of one or more embodiments of the photovoltaic modules disclosed in PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, entitled Building Integrated Photovoltaic System, and owned by GAF Energy LLC; and U.S. Pat. No. 11,251,744 to Bunea et al., issued Feb. 15, 2022, entitled “Photovoltaic Shingles and Methods of Installing Same,” and owned by GAF Energy, LLC, the disclosure and contents of each of which are incorporated by reference herein in their entirety.

(31) In some embodiments, the first side lap **206a** of the roofing shingle **200a** overlays the second side lap **208b** of the photovoltaic module **200b**. In some embodiments, the aperture **101** of the roof deck **100**, the aperture **210a** of the roofing shingle **200a**, and the aperture **212b** of the photovoltaic module **200b** are aligned with one another. In some embodiments, the aperture **101** of the roof deck **100**, the aperture **210a** of the roofing shingle **200a**, and the aperture **212b** of the photovoltaic module **200b** are substantially aligned with one another.

(32) In some embodiments, the passthrough **10** is installed on the first side lap **206a** of the roofing shingle **200a**. In some embodiments, the base **12** of the passthrough **10** is substantially parallel to the roof deck **100**. In some embodiments, a lower surface **15** of the base **12** is substantially parallel to the roof deck **100**. In some embodiments, the aperture **26** of the base **12** of the passthrough **10** is configured to align with the aperture **101** of the roof deck **100**, the aperture **210a** of the roofing shingle **200a**, and the aperture **212b** of the photovoltaic module **200b**.

(33) Referring to FIGS. 3A and 3B, in some embodiments, the first side lap **206a** of the roofing shingle **200a** and the second side lap **208b** of the photovoltaic module **200b** form a wireway **220**. In some embodiments, the passthrough **10** is located within the wireway **220** when the passthrough **10** is installed on the roof deck **100**. In some embodiments, a cover **222** is attached to the wireway **220**. In some embodiments, the cover **222** is removably attached to the wireway **220**. In some embodiments, the cover **222** covers the passthrough **10**. In some embodiments, the cover **222** inhibits water or moisture from entering the wireway **220**. In some embodiments, the cover **222** includes multiple cover portions or a plurality of the covers **222** connected to one another. In some embodiments, each of the wireway **220** and the cover **222** has a structure, composition, components, features, and/or function similar to those of one or more embodiments of the wireways and covers disclosed in PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, entitled Building Integrated Photovoltaic System, owned by GAF Energy LLC; and/or U.S. Pat. No. 11,251,744 to Bunea et al., issued Feb. 15, 2022, entitled “Photovoltaic Shingles and Methods of Installing Same,” and owned by GAF Energy, LLC, the disclosure and contents of each of which are incorporated by reference herein in their entirety.

(34) In some embodiments, an underlayment layer overlays the roof deck **100**. In some embodiments, the electrical cable passthrough **10** is installed on the underlayment layer. In some embodiments, a sealant is applied intermediate the underlayment layer and the base **12** of the electrical cable passthrough **10**. In some embodiments, the sealant includes butyl, silicone, rubber, epoxy, latex, neoprene, or polyurethane foam.

(35) In some embodiments, the interior portion **51** is sized and shaped to receive the electrical cables **120**. In some embodiments, the electrical cables **120** are connected to corresponding ones of the electrical connectors **122**. In some embodiments, the electrical cables **120** extend through the passthrough **10** and the conduit **110**. In some embodiments, the aperture **26** of the base **12** of the passthrough **10** is sized and shaped to receive the electrical cables **120**. In some embodiments, the first ends **121** of the electrical cables **120** extend outwardly from the first end **11** of base **12** and the

first end **42** of the lid **14** of the passthrough **10**. In some embodiments, each of the electrical cables **120** is a THHN stranded wire. In some embodiments, each of the electrical cables **120** is a XHHW stranded wire.

(36) Still referring to FIGS. **3A** and **3B**, in some embodiments, a receiver **150** includes a first end **152**, a second end **154** opposite the first end **152**, and an aperture **156** extending from the first end **152** to the second end **154**. In some embodiments, the receiver **150** has a symmetric shape. In some embodiments, the receiver **150** is tubular in shape. In some embodiments, the receiver **150** has a rectangular-tubular shape. In some embodiments, the receiver **150** has an asymmetric shape. In some embodiments, the receiver **150** has a curvilinear shape. In some embodiments, the receiver **150** has an elbow-shape.

(37) In some embodiments, the receiver **150** is installed underneath an inner surface **102** of the roof deck **100**. In some embodiments, the first end **152** of the receiver **150** is attached to the tubular portion **28** of the base **12** of the passthrough **10**. In some embodiments, the first end **152** of the receiver **150** is attached a fitting **158** which, in turn, is attached to the tubular portion **28** of the base **12** of the passthrough **10**. In some embodiments, the second end **154** of the receiver **150** is attached to an end of the conduit **110**. In some embodiments, the aperture **156** of the receiver **150** is sized and shaped to receive the at least one electrical cable **120** therethrough. In some embodiments, a rafter **104** is located underneath the inner surface **102** roof deck **100**. In some embodiments, the rafter **104** supports the roof deck **100**. In some embodiments, the rafter **104** is perpendicular to the roof deck **100**. In some embodiments, the rafter **104** is substantially perpendicular to the roof deck **100**. In some embodiments, the receiver **150** is installed proximate to a rafter **104**. In some embodiments, the receiver **150** is installed proximate to a first surface **106** of the rafter **104**. In some embodiments, the receiver **150** is installed adjacent to a first surface **106** of the rafter **104**. In some embodiments, the receiver **150** is configured to carry the at least one electrical cable **120** away from the rafter **104**. In some embodiments, the receiver **150** is configured to clamp the passthrough **10** to the roof deck **100**.

(38) In some embodiments, the receiver **150** extends 30 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 70 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 60 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 50 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 40 degrees relative to the inner surface **102** of the roof deck **100**.

(39) In some embodiments, the receiver **150** extends 40 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 40 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 40 degrees to 70 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 40 degrees to 60 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 40 degrees to 50 degrees relative to the inner surface **102** of the roof deck **100**. In some embodiments, the receiver **150** extends 50 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 50 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 50 degrees to 70 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 50 degrees to 60 degrees relative to the inner surface **102** of the roof deck **100**.

(40) In some embodiments, the receiver **150** extends 60 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 60 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 60 degrees to 70 degrees relative to the inner surface **102** of the roof deck **100**. In some

embodiments, the receiver **150** extends 70 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 70 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In some embodiments, the receiver **150** extends 80 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**.

(41) In some embodiments, a second one of the roofing shingle **200a** may be located in place of the photovoltaic module **200b**. In some embodiments, the roofing shingle **200a** may overlap at least a portion of another shingle **200a** that is vertically adjacent and below same. In some embodiments, at least a portion of the roofing shingle **200a** may be overlapped by yet another roofing shingle **200a** that is vertically adjacent and above same. In some embodiments, the roofing shingle **200a** may overlap at least a portion of another photovoltaic module **200b** that is vertically adjacent and below same. In some embodiments, at least a portion of the roofing shingle **200a** may be overlapped by another photovoltaic module **200b** that is vertically adjacent and above same.

(42) FIGS. **4** through **4D** show some embodiments of a passthrough **310** and components thereof installed on the roof deck **100**. In some embodiments, the passthrough **310** and the overall roofing system include a structure, function, components and features similar to those of the passthrough **10** and the corresponding system, with some differences. Referring to FIGS. **4A** and **4B**, in some embodiments, the passthrough **310** includes a base plate **311**, a sidewall **332** and a tubular member **328**.

(43) Referring to FIG. **4**, in some embodiments, the roofing system includes a first flashing member **312**. In some embodiments, the roofing system includes a second flashing member **314**. Referring to FIGS. **4**, **4C** and **4D**, in some embodiments, the first flashing member **312** includes a base **316**, an aperture **318** through the base **316**, and a sidewall **320** around at least a portion of a perimeter of the base **316**. In some embodiments, the aperture **318** is pre-formed in the base **318** during manufacturing thereof. In some embodiments, the base **316** may have one or more templates for a user to cut out material therefrom the form the aperture **318** for custom installation or alignment. In some embodiments, the first flashing member **312** includes a tab **315** extend from one end thereof.

(44) In some embodiment, the first flashing member **312** overlays and is above the first side lap **206a** of the roofing shingle **200a**. In some embodiments, a separate seal or sealing member is between the underside of the first flashing member **312** and the first side lap **206a** of the roofing shingle **200a**. In some embodiments, the seal is similar to the seal **111**. In some embodiments, the aperture **318** is aligned or substantially aligned with the aperture **210a** of the roofing shingle **200a**. In some embodiments, the passthrough **310** overlays and is above the first flashing member **312**. In some embodiments, the tubular member **328** of the passthrough **310** is installed within the aperture **318** of the first flashing member **312**. In some embodiments, the first flashing member **312** is configured to watershed water. In some embodiments, the sidewall **320** channels a flow of any water down the slope of the roof deck **100**. In some embodiments, the sidewall **320** diverts water away from the seal member. In some embodiments, another roofing shingle overlays the tab **315** of the first flashing member **312** to facilitate securement of the first flashing member **312** on the roof deck.

(45) In some embodiments, the second flashing member **314** is located between the roofing shingle **200a** and the photovoltaic module **200b**. In some embodiments, the second flashing member **314** is located between the lower surface of the first side lap **206a** of the roofing shingle **200a** and the upper surface of the photovoltaic module **200b**. In some embodiments, the second flashing member **314** is located between the lower surface of the first side lap **206a** of the first roofing shingle **200a** and the upper surface of the second side lap **208b** of the photovoltaic module **200b**. In some embodiments, the second flashing member **314** includes an aperture **322**. In some embodiments, the aperture **322** is aligned or substantially aligned with the aperture **318** of the first flashing member **312**, the aperture **210a** of the roofing shingle **200a**, the aperture **212b** of the photovoltaic module **200b**, and the aperture **101** of the roof deck **100**. In some embodiments, the second flashing

member **314** includes an upper edge **325**. In some embodiments, the upper edge **325** is angled relative to side portions thereof. In some embodiments, the second flashing member **314** is configured to seal to prevent water and/or watershed water. In some embodiments, a cover **360** covers the passthrough **310** and the first flashing member **312** to watershed water.

(46) In some embodiments, the tubular member **328** of the passthrough **310** is installed within and through the aperture **322** of the second flashing member **314**, the aperture **318** of the first flashing member **312**, the aperture **210a** of the roofing shingle **200a**, the aperture **212b** of the photovoltaic module **200b**, and the aperture **101** of the roof deck **100**. In some embodiments, a conduit **350** extends through the foregoing apertures and connects to the tubular member **328** of the passthrough **310** and houses and shrouds the electrical wires. In some embodiments, the upper edge **325** of the second flashing member **314** is configured to prevent any water that enters between the second side lap **208b** of the photovoltaic module **200b** and the underside of the second flashing member **314** from travelling laterally (left to right) and diverts and drains the water from right to left.

(47) Referring to FIGS. 5 and 6, in some embodiments, an electrical box **400** is installed underneath the roof deck **100** and within the structure. In some embodiment, the electrical box **400** is installed on the inner surface **102** of the roof deck **100**. In some embodiment, the electrical box **400** is installed proximate to the first surface **106** of the rafter **104**. In some embodiments, the electrical box **400** directly receives the tubular portion **28** of the passthrough **10**. In some embodiments, the conduit **110** is not utilized. In some embodiments, the electrical box **400** is a combiner box.

(48) In some embodiments, the passthrough **10** may include a structure, composition, function, and/or one or more features similar to those of more or one of the embodiments of the electrical cable passthrough devices disclosed in U.S. Pat. No. 11,177,639 to Nguyen et al., issued Nov. 16, 2021, entitled Electrical Cable Passthrough for Photovoltaic Systems, owned by GAF Energy LLC; and/or U.S. Pat. No. 11,658,470, to Nguyen et al. issued May 23, 2023, entitled Electrical Cable Passthrough for Photovoltaic Systems, owned by GAF Energy LLC, the disclosure and contents of each of which are incorporated by reference herein in their entirety.

(49) In some embodiments, the passthrough **10** is configured for use with electrical systems of building-integrated photovoltaic (BIPV) systems. In another embodiment, the passthrough **10** is configured for use with retrofit photovoltaic systems for roofing. In other embodiments, the passthrough **10** is configured for use with other electrical systems.

(50) The embodiments described herein are merely exemplary and that a person skilled in the art may make many variations and modifications without departing from the spirit and scope of the invention. All such variations and modifications are intended to be included within the scope of the invention.

Claims

1. A system, comprising: a roof deck; a passthrough installed on the roof deck, wherein the passthrough includes a base having a first end and a second end opposite the first end, a first surface extending from the first end to the second end, a second surface opposite the first surface, an aperture extending from the first surface to the second surface, and a plurality of first cavities, a lid attached to the base, wherein the lid includes a first end substantially aligned with the first end of the base, and a plurality of second cavities, a sidewall between the base and the lid, wherein the sidewall includes a first end, wherein the first end of the sidewall is proximate to the first end of the base, and a plurality of apertures, wherein the plurality of apertures of the sidewall is located at the first end of the sidewall, wherein each of the plurality of first cavities and each of a corresponding one of the plurality of second cavities are sized and shaped to form a corresponding one of the plurality of apertures, wherein the first surface is juxtaposed with the roof deck, and wherein the aperture of the base of the passthrough is configured to align with an aperture formed

within the roof deck; a plurality of cables, wherein each of the plurality of cables includes a first end and a second end opposite the first end of the cable, wherein each of the plurality of apertures of the sidewall is sized and shaped to receive a corresponding one of the plurality of cables; and a grommet installed on the at least one plurality of cables proximate to the first end of each of the plurality of cables, wherein the grommet is located between the plurality of first cavities and the plurality of second cavities, wherein the grommet includes a plurality of openings sized and shaped to receive a corresponding one of the plurality of cables, and a plurality of sections, wherein each of the plurality of openings is located in a corresponding one of the plurality of sections, wherein each of the plurality of sections of the grommet is located between a corresponding one of the plurality of first cavities and a corresponding one of the plurality of second cavities, wherein the lid and the base are configured to compress the grommet between the plurality of first cavities and the plurality of second cavities and around the plurality of cables when the lid is attached to the base, wherein the aperture of the base of the passthrough is sized and shaped to receive the plurality of cables, and wherein the first end of each of the plurality of cables extends outwardly from the first end of base and the first end of the lid of the passthrough.

2. The system of claim 1, wherein the lid is removably attached to the base.

3. The system of claim 1, further comprising a photovoltaic module installed on the roof deck, wherein the passthrough is installed proximate to the photovoltaic module.

4. The system of claim 1, wherein each of the plurality of cables includes an electrical connector, wherein each of the plurality of apertures of the sidewall is sized and shaped to receive a corresponding one of the electrical connector of the corresponding one of the plurality of cables.

5. The system of claim 1, wherein the base of the passthrough includes a tubular member extending from the first surface, wherein the tubular member includes the aperture of the base, and wherein the aperture of the roof deck is sized and shaped to receive the tubular member.

6. The system of claim 5, further comprising a receiver attached to the tubular member.

7. The system of claim 5, further comprising an electrical box installed underneath the roof deck, and wherein the electrical box directly receives the tubular member.

8. A cable passthrough, comprising: a base having a first end and a second end opposite the first end, a first surface extending from the first end to the second end, a second surface opposite the first surface, a plurality of first cavities, and an aperture extending from the first surface to the second surface; a lid attached to the base, wherein the lid includes a first end substantially aligned with the first end of the base, and a plurality of second cavities; and a sidewall between the base and the lid, wherein the sidewall includes a first end, wherein the first end of the sidewall is proximate to the first end of the base, and a plurality of apertures, wherein the plurality of apertures of the sidewall is located at the first end of the sidewall, wherein each of the plurality of first cavities and a corresponding one of each of the plurality of second cavities are sized and shaped to form a corresponding one of the plurality of apertures, wherein the cable passthrough is configured to be installed on a roof deck such that the first surface is juxtaposed with the roof deck, and wherein the aperture of the base is configured to align with an aperture formed within the roof deck, wherein the cable passthrough is configured to receive a plurality of cables, wherein the aperture of the base is sized and shaped to receive the plurality of cables, wherein the plurality of cables is capable of extending outwardly from the first end of the base and the first end of the lid of the cable passthrough, and wherein the lid and the base are configured to compress a grommet installed between the plurality of first cavities and the plurality of second cavities and around the plurality of cables when the lid is attached to the base, wherein the grommet includes a plurality of openings sized and shaped to receive a corresponding one of the plurality of cables, and a plurality of sections, wherein each of the plurality of openings is located in a corresponding one of the plurality of sections, wherein each of the plurality of sections is configured to be located between a corresponding one of the plurality of first cavities and a corresponding one of the plurality of second cavities.

9. A system, comprising: a passthrough installed on a roof deck, wherein the passthrough includes a base having a plurality of first cavities, a lid attached to the base, wherein the lid includes a plurality of second cavities, and a sidewall between the base and the lid, wherein the sidewall includes a plurality of apertures, wherein each of the plurality of first cavities and a corresponding one of each of the plurality of second cavities are sized and shaped to form a corresponding one of the plurality of apertures; a plurality of cables, wherein each of the plurality of apertures of the sidewall receives a corresponding one of the plurality of cables; and a grommet installed on the plurality of cables, wherein the grommet is located between the plurality of first cavities and the plurality of second cavities, wherein the grommet includes a plurality of sections, and a plurality of openings, each of which is located in a corresponding one of the plurality of sections, wherein each of the plurality of openings receives a corresponding one of the plurality of cables, wherein each of the plurality of sections of the grommet is located between a corresponding one of the plurality of first cavities and a corresponding one of the plurality of second cavities, and wherein the lid and the base are configured to compress the grommet between the plurality of first cavities and the plurality of second cavities and around the plurality of cables when the lid is attached to the base.
10. The system of claim 9, wherein the lid is removably attached to the base.
11. The system of claim 9, further comprising a photovoltaic module installed on the roof deck, wherein the passthrough is installed proximate to the photovoltaic module.
12. The system of claim 9, wherein each of the plurality of cables includes an electrical connector, wherein each of the plurality of apertures of the sidewall is sized and shaped to receive a corresponding one of the electrical connector of the plurality of cables.
13. The system of claim 9, wherein the base of the passthrough includes a tubular member, and wherein an aperture in the roof deck is sized and shaped to receive the tubular member.
14. The system of claim 13, further comprising a receiver attached to the tubular member.
15. The system of claim 13, further comprising an electrical box installed underneath the roof deck, and wherein the electrical box directly receives the tubular member.
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